



JIANGNAN HUANG

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Address: Nijmegen, The Netherlands (Open to relocation)

Languages: English (C1); French (B2); Dutch (A1); Chinese (Native)

[Website](#) | [Scholar](#) | [GitHub](#) | [LinkedIn](#)

EXPERIENCE

Ph.D. Researcher - AI Workflow Automation & Software Reliability

Oct. 2022 – Present

iCIS, Radboud University

Nijmegen, The Netherlands

- Built FLOWSTEP, an LLM-supported workflow automation system in Python that converts partial CI/CD context into executable next-step recommendations; improved completion accuracy by 33.6%.
- Engineered data pipelines over large GitHub Actions ecosystems by parsing YAML, normalizing noisy metadata, deriving SQL-ready benchmark tables, and documenting assumptions for repeatable analysis.
- Developed CIGAR, a retrieval-based recommender over indexed automation metadata with candidate scoring, context matching, and top-k evaluation; reached 95.5% top-5 accuracy.
- Investigated software-supply-chain risks in automation workflows, identifying 33K+ security issues while training outcome-prediction models with 90% F1.
- Created evaluation harnesses with baselines, regression checks, and failure analysis so automation recommendations could be reviewed by technical stakeholders.

Machine Learning Scientist

Oct. 2021 – Mar. 2022

Huawei Technologies France

Boulogne-Billancourt, France

- Modelled virtualized cloud scheduling as a reinforcement-learning problem, evaluating Python experiments under scalability, latency, and resource-efficiency constraints.

Machine Learning Scientist (Internship)

Mar. 2021 – Aug. 2021

Inria Saclay Center

Palaiseau, France

- Evaluated privacy and utility trade-offs in ML models, relating empirical performance to governance constraints for sensitive-data deployment settings.

EDUCATION

Ph.D. in Computer Science (AI for Software Engineering)

Oct. 2022 – Expected Oct. 2026

iCIS, Radboud University. Advised by Dr. Bin Lin & Prof. Dr. Frits Vaandrager

Nijmegen, The Netherlands

Teaching: Software Engineering; Software Verification.

MSc in Computer Science (AI/ML)

Sept. 2019 – Aug. 2021

Université Paris-Saclay. Advised by Prof. Dr. Isabelle Guyon

Orsay, France

BEng in Opto-electronic Information Science and Engineering

Sept. 2014 – June 2018

Huazhong University of Science and Technology (HUST)

Wuhan, China

SELECTED PUBLICATIONS

- FLOWSTEP: Framework for Learning to Predict GitHub Actions Workflow Steps
Jiangnan Huang, Bin Lin, and Yutaro Kashiwa
Under Review
- On the Reruns of GitHub Actions Workflows
Jiangnan Huang and Bin Lin
TOSEM 2026  
- Revisiting Security Practices for GitHub Actions Workflows
Jiangnan Huang and Bin Lin
ICPC 2025  
- CIGAR: Contrastive Learning for GitHub Action Recommendation
Jiangnan Huang and Bin Lin
SCAM 2023 Best Paper  

SKILLS

AI & ML: PyTorch; Transformers; embeddings; representation learning; RAG; AI agents; LLM evaluation

Data & Retrieval: SQL; Spark; pandas; NumPy; information retrieval; unstructured-data processing

Engineering: Python; Java; FastAPI; REST APIs; data/ML pipelines; Git; DevOps; CI/CD; Docker; AWS/Azure

Research: empirical research; model benchmarking; experiment design; statistical analysis; reproducible pipelines

Collaboration: LaTeX; design documentation; stakeholder communication; project ownership; agile mindset

HOBBIES

Sports: Avid outdoor sports enthusiast: bouldering, snowboarding, free-diving, tennis, and more.

Photography: Active street photographer, with a focus on urban life, travel, and candid portrait work.